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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE

Course Code & Title: CS3452 – Theory of Computation

Name of the Faculty Member(s): Mrs. S. Manjula

Innovative Practice Description

Unit / Topic: Unit II / Equivalence of Finite Automata and Regular Expressions

Course Outcome: CO 2

Topic Learning Outcome: TLO 5

Activity Chosen: Regex Automata Matching

Date of Activity: 27.01.2026

Justification:

- The "Regex Automata Matching" activity was chosen because students often find it challenging to understand the relationship between Regular Expressions and Finite Automata, despite their fundamental importance in Theory of Computation.
- The activity provides a hands-on and collaborative learning environment where students actively construct and match regular expressions with corresponding automata, helping them visualize abstract concepts. By engaging in group discussions and problem-solving, students gain a deeper understanding of language recognition, automata design, and pattern matching. Furthermore, the activity highlights the practical applications of these concepts in compiler construction, lexical analysis, and text-processing tools, thereby bridging the gap between theory and real-world computing systems.

Time Allotted for the Activity: 30 Minutes

Details of the Implementation:

The activity leverages the Kleene's theorem, which establishes the formal equivalence between the expressive power of regular expressions and finite automata. Students were required to perform bidirectional conversions and verify the equivalence by testing the constructed automaton against strings generated by the original regular expression.

Step 1: Theoretical Briefing (10 minutes)

- Faculty provided a concise review of regular expression operators: union ($\cup$), concatenation ($\circ$), Kleene star ($*$), and positive closure ($+$).
- The formal definition of an NFA and DFA were recalled: $M = (Q, \Sigma, \delta, q_0, F)$.

- Thompson's Construction algorithm was outlined on the board: decomposing a regex into primitive NFAs for ϵ , single symbol, union, concatenation, and Kleene star, and composing them with ϵ -transitions.
- The subset construction (powerset) algorithm for NFA-to-DFA conversion and the state-elimination method for DFA-to-RE conversion were briefly reviewed.

Step 2: Problem Distribution to Groups (5 minutes)

- Each group of 3–4 students received a printed worksheet containing: (a) a regular expression, (b) a language description in set-builder notation, and (c) a set of test strings to classify as accepted or rejected.
- Problems included:
 - $bc(ab + c)^*a^+$
 - $(ab + c)^*$
 - $(a|b)^*abb$
 - $10+(0+11)0^*1$
 - $(01+1)^*$

Step 3: RE to FA Construction (10 minutes)

- Students applied Thompson's Construction systematically: each atomic sub-expression was drawn as a two-state NFA; union, concatenation, and Kleene star operations were built up using ϵ -transitions following the standard structural induction.
- All states were labelled $q_0, q_1, \dots$ and all transitions including ϵ -moves were explicitly annotated on the worksheets.
- Groups identified the unique accepting state q_f of the resulting NFA.

Step 4: Verification and Discussion (5 minutes)

- Each group tested their FA against the provided test strings, tracing execution paths state by state.
- Groups cross-checked results with adjacent teams; discrepancies were analysed to identify errors in ϵ -closure computation or transition table construction.
- A class-wide discussion consolidated understanding of why the two formalisms are equally expressive despite their structural differences.

CO – PO / PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
CO 1	2	2	2	2	1	1	1

1–Low 2–Moderate 3–High

PO / PSO Justification:

PO / PSO	Justification
PO1	Students applied discrete mathematics and formal language theory to construct regular expressions for given languages and derive equivalent finite automata, directly exercising foundational engineering mathematics.

PO2	By expressing complex language-recognition problems as regular expressions and equivalent finite automata, students practised formulating abstract computational problems in a standard engineering notation.
PO3	Students designed regular expressions to describe languages recognised by lexical analyzers, reinforcing the direct application of formal language theory in compiler design and other system-software components.
PO4	Groups designed and analysed system inputs by constructing regular expressions and verifying them against corresponding DFA/NFA models, exercising systematic design-analysis methodology.
PO9	Students worked together to discuss different regex and automata solutions, resolve disagreements, and arrive at a correct common answer, improving their teamwork and communication skills.
PO10	Students read problem statements expressed in natural language and mathematical notation, then interpreted and communicated equivalent formal representations in both regex and automaton form.
PSO1	Students will be able to apply the concepts of Regular Expressions and Finite Automata learned through this activity to develop lexical analyzers and other components used in compiler design.

Images / Screenshot of the Practice:



Figure:1 Students engaged in collaborative Regex-to-FA construction and verification

Reflective Critique:**Feedback of Practice from Students and Other Stakeholders:**

- Students responded positively to the activity and appreciated the interactive group-based learning approach.
- The hands-on matching of Regular Expressions with Finite Automata helped students better understand the relationship between the two concepts.
- Students reported increased confidence in constructing automata and identifying equivalent regular expressions.
- Faculty observers noted active participation and meaningful discussions among students during the activity.

Benefit of the Practice:

- Improved students' understanding of Regular Expressions, Finite Automata, and their equivalence.
- Enhanced problem-solving, analytical thinking, and collaborative learning skills through group discussions.

Challenges Faced in Implementation:

- Some students initially found it difficult to convert Regular Expressions into corresponding automata and vice versa.
- A few groups required additional guidance to verify the correctness of their solutions.

References:

1. <https://www.ishcmc.com/news-and-blog/peer-led-learning/>
2. <https://www.cascadiaschool.org/what-are-the-benefits-of-mixing-ages-and-grades-in-a-classroom-doesnt-this-make-teaching-way-harder-sorryteachers>
3. <https://www.21kschool.com/in/blog/benefits-of-activity-based-learning/>
4. <https://www.edutopia.org/topic/collaborative-learning>
5. https://www.ritrjpm.ac.in/images/computer-science/28.CS8501_ColloborativeLearning.pdf

Signature of Faculty Member

HOD / CSE